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functions:

A function is a block of reusable code that performs a specific task.

it helps in breaking a large program into smaller manageable parts.

function denoted by fun_name()

why functions?

* increases code reusability.
* improves readability and modularity.
* easier debugging and testing.

Types of function

1. built in functions: provides by python (print(), input(), len(), sum()...etc)
2. user defined functions: created by the programmer/developer using def (define)

Syntax:

```
def function_name(parameters):
        statements
        return value
```

* function without input and without return
* function without input and with return
* function with input and without return
* function with input and with return.

3. lambda function:

anonymous, single line functions are called created with lambda.

local variable : local variable is declared outside the function but cannot access outside the function unless function is returning.

global variable : global variable declared outside the function and use anywhere in that file.

* function without input and without return.

Syntax :

```
def fun_name():
    statements
# call the function
fun_name()
```

* function without input and with return

Syntax :

```
def fun_name():
    statements
    return value
# call the function
v1, v2, ... vn = function_name()
```

* function with input and without return.

```
def fun_name(p1, p2, ... pn):
    statements
# direct calling
fun_name(v1, v2, ... vn)
# based on user input
v1 = input()
v2 = input()
...
vn = input()
fun_name(v1, v2, ... vn)
```

* function with input and with return

```
def fun_name (P1, P2, .. Pn):
        Statements
        return values
# call the fuc
# func_name (V1, V2 .. Vn)
# based on user input
V1 = input()
V2 = input()
...
Vn = input()
function_name (V1, V2, ... Vn)
```